

LARGE SAMPLE MAXIMUM LIKELIHOOD ESTIMATION IN A NORMAL-LOGNORMAL DISTRIBUTION

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ABSTRACT

The joint distribution of random variables X and Y is examined for which the marginal distribution of X is normal, the conditional distribution of Y given X is lognormal and the conditional expectation of Y given X is of an exponential form. In one situation the maximum likelihood equations for estimating the parameters cannot be solved explicitly for one of the parameters, but a solution for that parameter may be obtained iteratively and the whole set of equations solved. A numerical example is considered and the results of the estimation of the parameters by maximum likelihood is compared to the results obtained by the method of moments.

1. Introduction

Consider a bivariate distribution of X and Y which meets the following three assumptions:

- (1) The marginal distribution of X is normal with mean μ_1 and variance σ_1^2
- (2) The conditional distribution of Y given X is lognormal with parameters $\mu_2(x)$, σ_2^2 , and a . That is

$$f_{2.1}(y|x) = \frac{1}{\sqrt{2\pi}\sigma_2(y-a)} \exp - \frac{1}{2} \left[\frac{\ln(y-a) - \mu_2}{\sigma_2} \right]^2, y > a.$$

- (3) $E(Y|X=x) = \exp(\gamma x + \beta) + a$.

Throughout this paper X and Y will be considered random variables with specific realizations x and y . The regression parameters γ and β are, in general, unknown and one of the problems treated in this paper is their estimation.

Using these three assumptions one obtains the bivariate density function

$$f(x,y) = \frac{1}{2\pi\sigma_1\sigma_2(y-a)} \exp - \frac{1}{2} \left[\frac{(x-\mu_1)^2}{\sigma_1^2} + \frac{(\ln(y-a) - (\gamma x + \beta - \frac{1}{2}\sigma_2^2))^2}{\sigma_2^2} \right]$$

$$-\infty < x < \infty, y > a.$$

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Yuan (1933) obtained estimators for the parameters of this distribution, and for the bivariate lognormal distribution by the method of moments; another approach to the bivariate normal case was taken by Mostafa and Mahmoud (1964). This article will consider estimation by maximum likelihood methods.

Contours of constant density for various values of β , γ , and σ_1^2 with $\mu_1 = 0$ and $\sigma_1^2 = 225$ are shown in Figures 1.1 through 1.3. The quantities μ_1 and σ_1^2 are held constant since changing μ_1 only shifts the horizontal axis and changing σ_1^2 only changes the horizontal spread. Note that the figures can be used to illustrate the effects of changing only one parameter at a time. Scales have not been included on the graphs nor are the levels of the contours indicated since the graphs would not be used to read numerical values, but would be used only to illustrate the shape of the surface and to compare the effects of parameter changes. To this end the scales and levels of the contours are the same from graph to graph.

It should be noted that different assumptions concerning the form of the regression function will lead to different bivariate normal-lognormal distributions.

If one makes the transformation $Z = \ln Y$, it is seen that X and Z have a bivariate normal distribution with $E(X) = \mu_1$, $E(Z) = \gamma\mu_1 + \beta - \frac{1}{2}\sigma_2^2$ and covariance matrix

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \gamma\sigma_1^2 \\ \gamma\sigma_1^2 & \gamma^2\sigma_1^2 + \sigma_2^2 \end{pmatrix}.$$

The above implies that Y is lognormally distributed with

$$E(Y) = \exp \left[\gamma\mu_1 + \beta + \frac{\gamma^2\sigma_1^2}{2} \right]$$

$$\text{Var}(Y) = \exp [2\gamma\mu_1 + 2\beta + \gamma^2\sigma_1^2] [\exp(\gamma^2\sigma_1^2 + \sigma_2^2) - 1].$$

We also see that the conditional distribution of X given Y is normal with

$$E(X|Y=y) = \frac{\mu_1\sigma_2^2 - \sigma_1^2\gamma(\beta - \frac{\sigma_2^2}{2} - \ln y)}{\gamma^2\sigma_1^2 + \sigma_2^2}$$

$$\text{Var}(X|Y=y) = \frac{\sigma_1^2\sigma_2^2}{\gamma^2\sigma_1^2 + \sigma_2^2}.$$

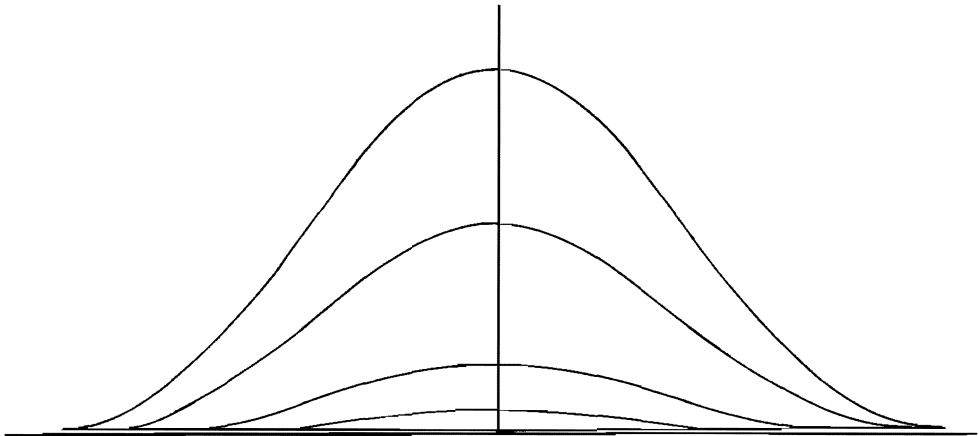


Figure 1.1
 $\beta = 10; \gamma = 0; \sigma_2^2 = 5.$

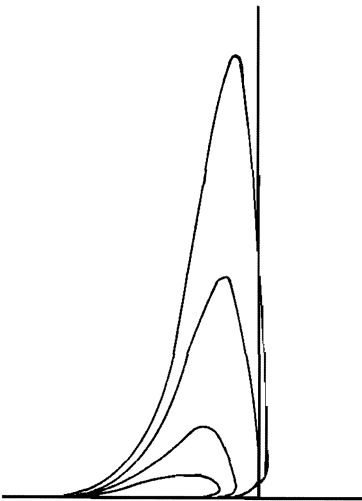


Figure 1.2
 $\beta = 13,5; \gamma = 0,75; \sigma_2^2 = 5$

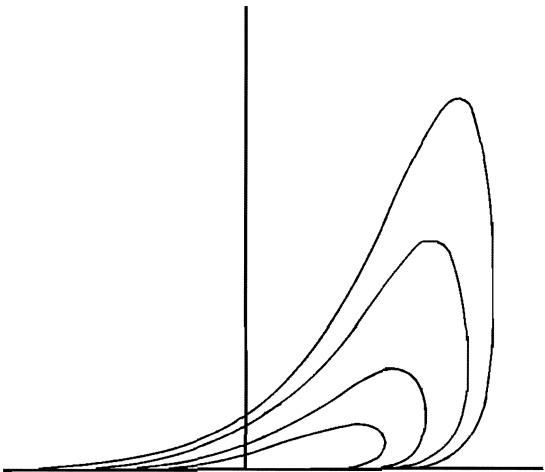


Figure 1.3
 $\beta = 5; \gamma = 0,2; \sigma_2^2 = 1$

It may be seen by direct evaluation that

$$\text{Cov}(X, Y) = \gamma \sigma_1^2 \exp[\gamma \mu_1 + \beta + \frac{1}{2} \gamma^2 \sigma_1^2]$$

The ratio of the correlation coefficient $\rho_{X,Y}$ to $\rho_{X, \ln Y}$ is

$$\begin{aligned} \frac{\rho_{X,Y}}{\rho_{X, \ln Y}} &= \frac{\frac{\gamma \sigma_1}{\sqrt{\exp[\gamma^2 \sigma_1^2 + \sigma_2^2] - 1}}}{\frac{\gamma \sigma_1}{\sqrt{\gamma^2 \sigma_1^2 + \sigma_2^2}}} \\ &= \sqrt{\frac{\gamma^2 \sigma_1^2 + \sigma_2^2}{\exp[\gamma^2 \sigma_1^2 + \sigma_2^2] - 1}} \end{aligned}$$

This ratio exhibits some interesting properties, first

$$|\rho_{X,Y}| \leq |\rho_{X, \ln Y}|$$

with equality only for $\gamma = 0$, in which case both correlations are zero. Secondly, for large values of $\gamma^2 \sigma_1^2 + \sigma_2^2$, high correlation between X and $\ln Y$ does not imply high correlation between X and Y . Further, if the quantities σ_1^2 , σ_2^2 , and $\gamma^2 \sigma_1^2 + \sigma_2^2$ are small then

$$\frac{\rho_{x,y}}{\rho_{x, \ln y}} \rightarrow 1$$

2. Estimation of Parameters

If a is known, maximum likelihood estimation of the parameters is easily accomplished by working with the re-parameterized bivariate normal distribution.

A rather interesting result occurs when a , μ_1 , and σ_1^2 are known. One obtains maximum likelihood estimators

$$\hat{\gamma} = \frac{(x_i - \bar{x}) \ln(y_i - a)}{(x_i - \bar{x})^2}$$

$$\hat{\beta} = \frac{1}{n} \sum \ln(y_i - a) + \frac{\sigma_2^2}{2} - \gamma \bar{x}$$

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These estimators are unbiased with variance covariance matrix

$$\Sigma_{\hat{\gamma}, \hat{\beta}} = \begin{vmatrix} \frac{\sigma_2^2}{(n-3)\sigma_1^2} - \frac{\sigma_2^2 \mu_1}{(n-3)\sigma_1^2} & \\ -\frac{\sigma_2^2 \mu_1}{(n-3)\sigma_1^2} & \frac{\sigma_2^2}{n} + \frac{\sigma_2^2}{(n-3)\sigma_1^2} \left[\frac{\sigma_1^2}{n} + \mu_1^2 \right] \end{vmatrix}$$

Now let us investigate the possibility of obtaining more efficient estimators of γ and β by the use of μ_1 and σ_1^2 rather than their estimators $\bar{x}$ and s_x^2 . To this purpose construct the unbiased estimators

$$\hat{\gamma} = \frac{\sum (x_i - \mu_1) \ln(y_i - a)}{n\sigma_1^2}$$

and

$$\hat{\beta} = \frac{1}{n} \sum \ln(y_i - a) + \frac{\sigma_2^2}{2} - \hat{\gamma}\mu_1$$

Computation of the relative efficiency of $\tilde{\gamma}$ to $\hat{\gamma}$ shows $\tilde{\gamma}$ to be a more efficient estimator when

$$2\gamma^2\sigma_1^2 + \left(\gamma\mu_1 + \beta - \frac{\sigma_2^2}{2} \right)^2 < \frac{3\sigma_2^2}{n-3}$$

Data were generated for various values of n for which the above inequality held and little advantage was seen for $\tilde{\gamma}$; also, relative to the Cramer-Rao lower bound, $\hat{\gamma}$ is asymptotically efficient while $\tilde{\gamma}$ is always inefficient unless $\gamma = 0$ and $\beta = \frac{\sigma_2^2}{2}$. All of these

facts lead to the choice of $\hat{\gamma}$ as a generally better estimator than $\tilde{\gamma}$. Similar arguments show that, in general, $\hat{\beta}$ is better than $\tilde{\beta}$.

Consider now the more general case in which all six parameters are to be

estimated. In this situation, the maximum likelihood estimator for μ_1 is $\bar{x}$ while that for σ_1^2 is $s_x'^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$. The usual adjustment for bias is made on the latter to obtain $s_x^2 = \frac{n}{n-1} s_x'^2$. Maximum likelihood estimators for γ , β , σ_2^2 , and α are found numerically using a method proposed by Cohen (1951). Aitchison and Brown (1957), Lambert (1964) and Hill (1963) have all stated that maximum likelihood estimation is open to fundamental criticism. However, Harter and Moore (1966) have pointed out that for reasonable sample sizes maximum likelihood estimates, although questionable from a theoretical standpoint, give good estimates of the parameters. A numerical illustration is given in the next section.

If L is the likelihood function, the problem is to solve the simultaneous system of equations

$$\begin{aligned} \frac{\partial \ln L}{\partial \gamma} &= 0, & \frac{\partial \ln L}{\partial \sigma_2^2} &= 0, \\ \frac{\partial \ln L}{\partial \beta} &= 0, & \frac{\partial \ln L}{\partial \alpha} &= 0, \end{aligned}$$

The first three equations yield

$$\begin{aligned} \hat{\gamma} &= \frac{\sum (x_i - \bar{x}) \ln (y_i - \hat{a})}{\sum (x_i - \bar{x})^2} \\ \hat{\sigma}_2^2 &= \frac{1}{n} \left[\sum \ln^2 (y_i - \hat{a}) - \frac{1}{n} \left(\sum \ln (y_i - \hat{a}) \right)^2 - \hat{\gamma}^2 \sum (x_i - \bar{x})^2 \right] \\ \hat{\beta} &= \frac{1}{n} \sum \ln (y_i - \hat{a}) - \hat{\gamma} \bar{x} + \frac{\hat{\sigma}_2^2}{2} \end{aligned}$$

Substitution of these relations into $\frac{\partial \ln L}{\partial \alpha} = 0$ yields $\lambda(\alpha) = 0$

$$\begin{aligned} \text{where } \lambda(\hat{a}) &= \frac{1}{n} \sum \frac{1}{y_i - \hat{a}} \left[\sum \ln^2 (y_i - \hat{a}) - \frac{1}{n} \left\{ \sum \ln (y_i - \hat{a}) \right\}^2 \right. \\ &\quad \left. - \frac{\left\{ \sum (x_i - \bar{x}) \ln (y_i - \hat{a}) \right\}^2}{\sum (x_i - \bar{x})^2} - \sum \ln (y_i - \hat{a}) \right] \\ &\quad + \sum \frac{\ln (y_i - \hat{a})}{y_i - \hat{a}} - \frac{\sum (x_i - \bar{x}) \ln (y_i - \hat{a})}{\sum (x_i - \bar{x})^2} - \sum \frac{x_i - \bar{x}}{y_i - \hat{a}}. \end{aligned}$$

Once $\hat{\alpha}$ has been found, $\hat{\gamma}$, $\hat{\sigma}_2^2$, and $\hat{\beta}$ are readily computed. A FORTRAN program for computation of these estimators which solves for $\hat{\alpha}$ by the method of false position has been prepared by W.R. Schucany of the S.M.U. Statistical Laboratory and is to be found in Crofts (1969).

In practice, the above method seems to break down for large values of σ_2^2 . The asymptotic normality of the maximum likelihood estimators is not valid for $\sigma_2^2 \geq \frac{1}{3}$ and

this may be the critical point for the existence of a solution to $\lambda(\hat{\alpha}) = 0$, although the authors have not yet determined whether this is the case. This restriction on σ_2^2 may not be a source of difficulty in many practical situations. Recall that the variance of $\ln Y$, which is normally distributed, is given by $\gamma^2 \sigma_1^2 + \sigma_2^2$. Aitchison and Brown (1957) have pointed out that frequently this is around 0.5. Thus if the variables are not independent and the variance of X is of reasonable magnitude, we would expect σ_2^2 to be less than one third.

Although it is not possible to check these estimators for bias, if $\sigma_2^2 < \frac{1}{3}$, they would be consistent and asymptotically normally distributed with the following large-sample variances and covariances

$$\text{Var}(\bar{x}) = \frac{\sigma_1^2}{n}$$

$$\text{Var}(\hat{\sigma}_1^2) = \frac{2\sigma_1^4}{n}$$

$$\text{Var}(\hat{\gamma}) = \frac{1}{2nc} (b - e^{2a})$$

$$\text{Var}(\hat{\sigma}_2^2) = \frac{\sigma_1^2 \sigma_2^2}{nc} \{b - e^{2a} (1 + \gamma^2 \mu_1^2)\}$$

$$\text{Var}(\hat{\beta}) = \frac{1}{nc} \left\{ \sigma_1^2 \left(\frac{\sigma_2^2 + 2}{4} \right) + \frac{\mu_1^2}{2} b \right\}$$

$$- \frac{1}{4} e^{2a} [(\sigma_1^2 + \mu_1 \gamma) \sigma_2^2 + 2\mu_1 (\mu_1 - \gamma \sigma_1^2)]$$

$$- \gamma \sigma_1^2 (\mu_1 - \gamma \sigma_1^2) (\sigma_2^2 + 2)] \}$$

$$\text{Var}(\hat{a}) = \frac{\sigma_1^2}{2nc}$$

$$\text{Cov}(\bar{x}, s_x^2) = \text{Cov}(\bar{x}, \hat{\gamma}) = \text{Cov}(\bar{x}, \sigma_2^2) = \text{Cov}(\bar{x}, \hat{\beta}) = \text{Cov}(\bar{x}, \hat{a}) = 0$$

$$\text{Cov}(s_x^2, \hat{\gamma}) = \text{Cov}(s_x^2, \hat{\sigma}_2^2) = \text{Cov}(s_x^2, \hat{\beta}) = \text{Cov}(s_x^2, \hat{a}) = 0$$

$$\text{Cov}(\hat{\gamma}, \hat{a}_2^2) = 0$$

$$\text{Cov}(\hat{\gamma}, \hat{\beta}) = \frac{1}{2nc} [\mu_1 b - e^{2a} (\mu_1 - \gamma \sigma_1^2)]$$

$$\text{Cov}(\hat{\gamma}, \hat{a}) = \frac{\gamma \sigma_1^2 e^a}{2nc}$$

$$\text{Cov}(\hat{\sigma}_2^2, \hat{\beta}) = \frac{\sigma_1^2 \sigma_2^2}{2nc} \{ b - e^{2a} (1 + \gamma^2 \sigma_1^2) \}$$

$$\text{Cov}(\hat{\sigma}_2^2, \hat{\beta}) = 0$$

$$\text{Cov}(\hat{\beta}, \hat{a}) = \frac{1}{2nc} e^a \sigma_1^2 (1 + \gamma)$$

where

$$a = \sigma_2^2 - \beta - \gamma \mu_1 + \frac{\gamma^2}{2} \sigma_1^2$$

$$b = (1 - 3\sigma_2^2) e^{2a} + \gamma^2 \sigma_1^2 + \sigma_2^2$$

$$c = \frac{\sigma_1^2}{2\sigma_2^2} \{ b - e^{2a} (1 + \gamma^2 \sigma_1^2) \}$$

3. Numerical Example

Table 1 gives the heights and weights of 11 382 school boys. This data, which was given by Isserlis (1915), is considered by Yuan (1933) to be a sample from the bivariate distribution under consideration, with height being approximately normally distributed and weight having approximately a lognormal distribution. Yuan estimated the parameters by the method of moments; a comparison of these estimates with the maximum likelihood estimates is given in Table 2.

Chi-square goodness of fit tests were applied to the data with the following

results:

$$\chi^2_{\text{Method of Moments}} = 8637,34$$

$$\chi^2_{\text{Maximum Likelihood}} = 785,18$$

These results show a considerable advantage for the estimates obtained by maximum likelihood methods.

TABLE 1
Heights and Weights of School Boys

	Height in Inches (x)												Total
	30 32	33 35	36 38	39 41	42 44	45 47	48 50	51 53	54 56	57 59	60 62	63 65	
24-28	4	9	2			1							16
29-33	3	42	62	25	1	1							136
34-38		16	220	414	72	6							728
39-43	1	3	51	617	697	95	11	1					1476
44-48		1	7	122	875	603	38	8	1				1655
49-53			4	12	249	988	411	33	5	4			1706
54-58		1	3	1	17	436	905	171	11	4	3		1552
59-63			1		1	39	630	568	51	6	1		1297
64-68				1		8	161	621	206	3	2	2	1004
69-73				1			35	374	340	24	2		776
74-78							3	106	335	76	5		525
79-83							2	22	120	93	4	1	242
84-88						1		8	32	87	8	2	138
89-93								1	10	36	18	1	66
94-98									3	23	9	2	37
99-103										5	11	3	19
104-108									1		5	1	7
109-113											1		1
114-118													0
119-123												1	1
Total	8	72	350	1193	1914	2178	2196	1913	1115	361	69	13	382

TABLE 1

Comparison of Values of Parameters Obtained by the
Method of Moments with those Obtained by
Maximum Likelihood Estimation

Parameter	Method of Moments	Maximum Likelihood
μ_1	47,4644	47,4644
σ^2	27,3456	28,7275
a_1	-11,25	-4,1034
γ	0,02146	0,03712
σ^2	0,02462	0,007376
β	3,1705	2,2970

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